Computer Networks

Amazon SDE-1 / FAANG Interview Master Notes

A complete interview-oriented companion to Kunal Kushwaha's Computer Networking Full Course, with additional topics commonly tested in software-engineering interviews.

Primary source course: Kunal Kushwaha — Computer Networking Full Course: OSI Model Deep Dive with Real Life Examples

Video length: approximately 4 hours 7 minutes

How to use this PDF

- First pass: read the explanations and diagrams; do not memorize blindly.
- Second pass: answer the interview questions aloud without looking.
- Third pass: revise the final cheat sheets and the URL-request journey.
- For Amazon SDE-1, prioritize TCP/UDP, DNS, HTTP/HTTPS, OSI/TCP-IP, IP addressing, routing, NAT, sockets, MTU/MSS, flow/congestion control, and troubleshooting.

> **Important:** These are original study notes and explanations, not a transcript of the video. Extra sections are explicitly marked as FAANG/Amazon additions.

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1. Networking Fundamentals

What is a computer network?

A computer network is a collection of interconnected devices that exchange data using agreed protocols. The devices may include laptops, phones, servers, routers, switches, printers, IoT devices and cloud infrastructure.

Why networking matters to an SDE

Almost every backend or distributed application depends on a network: API calls, database connections, microservices, authentication, queues, caches, object storage and service-to-service communication.

Internet vs Web

The Internet is the underlying global network of interconnected networks. The Web is one service that runs over it using protocols such as HTTP/HTTPS.

Protocol

A protocol is a set of rules for communication. Examples include IP, TCP, UDP, DNS, HTTP, TLS and SMTP.

> Example: When a React frontend calls https://api.example.com/users, DNS may resolve the hostname, TCP or QUIC provides transport, TLS protects the connection, and HTTP carries the API request.

2. Client–Server, P2P and Network Architecture

Client–Server

The client initiates a request; the server provides a service. A browser is a client, while a web server is a server. A single machine can be a client in one interaction and a server in another.

Example: Browser → GET /products → API server → database → JSON response.

Peer-to-Peer

In P2P systems, peers can both consume and provide resources. This can distribute load and reduce dependence on one central server.

Important interview distinction

Concept	Question it answers
Client/server	Who requests and who serves?
Protocol	What rules are used?
IP	Which network endpoint?
Port	Which service/process endpoint?
Socket	Which communication endpoint/association?

3. IP Addresses and Ports

IPv4

IPv4 addresses are 32 bits, normally written as four decimal octets, e.g. 192.168.1.10. Each octet ranges from 0 to 255. The theoretical address space is 2^32 ≈ 4.29 billion.

Port numbers

A port identifies a transport-layer service endpoint on a host. Port numbers are 16 bits, so the range is 0–65535.

Port	Typical service
20/21	FTP data/control
22	SSH
25	SMTP
53	DNS
67/68	DHCP server/client
80	HTTP
110	POP3
143	IMAP
443	HTTPS

Ephemeral ports

A client usually uses a temporary source port for an outgoing connection. A simplified TCP connection can be described by source IP:source port and destination IP:destination port, plus the protocol.

`Example: 192.168.1.10:53124 → 142.250.1.1:443.`

4. LAN, MAN, WAN and Physical Internet

LAN

Local Area Network: home, office, lab or other local scope. Ethernet and Wi■Fi are common LAN technologies.

MAN

Metropolitan Area Network: spans a city or metropolitan region.

WAN

Wide Area Network: connects geographically distant networks. The Internet is a global interconnection of many networks.

Submarine cables

Intercontinental Internet traffic often travels over fiber-optic submarine cables. Wireless access from your laptop does not mean the whole Internet is wireless.

Modem vs router

A modem terminates/translates the access technology used by an ISP; a router forwards traffic between networks and commonly provides NAT, DHCP and firewall functions in home equipment. Exact device responsibilities vary by architecture.

5. Network Topologies

Topology	Idea	Typical trade-off
Bus	Shared backbone	Simple but backbone failure/ contention issues
Ring	Each node connects in a ring	Path dependency; variants add redundancy
Star	Nodes connect to a central device	Easy management; central device matters
Tree	Hierarchical stars	Scalable hierarchy; upper-level failure affects branches
Mesh	Many interconnections	High redundancy; expensive/complex

6. OSI Model — All 7 Layers

Mnemonic from top to bottom: Application → Presentation → Session → Transport → Network → Data Link → Physical.

Layer	Name	Core responsibility	Typical examples
7	Application	Application-level network services	HTTP, DNS, SMTP, FTP, SSH
6	Presentation	Representation, encoding, compression, encryption concepts	JSON, encoding, TLS-related functions
5	Session	Session establishment/management concepts	RPC/session mechanisms
4	Transport	Process-to-process delivery, ports, reliability/flow where provided	TCP, UDP
3	Network	Logical addressing and routing	IP, routers
2	Data Link	Local delivery, framing, MAC addressing	Ethernet, Wi■Fi, switches
1	Physical	Bits/signals over medium	Fiber, copper, radio

Why layers?

Layering isolates responsibilities. An application developer should not need to implement electrical signaling to make an HTTP request. A failure can also be reasoned about layer by layer.

PDU names

A common interview shorthand is: Application = data/message; TCP = segment; UDP = datagram; IP = packet; Ethernet/Wi■Fi = frame; Physical = bits.

> Important nuance: real protocol stacks do not always map perfectly to the OSI textbook boundaries.

7. TCP/IP Model and OSI vs TCP/IP

TCP/IP-style 5-layer view	OSI layers	Examples
Application	7, 6, 5	HTTP, DNS, SMTP
Transport	4	TCP, UDP
Network/Internet	3	IP, ICMP
Data Link	2	Ethernet, Wi■Fi
Physical	1	Signals/media

OSI vs TCP/IP

OSI is a conceptual reference model. TCP/IP is the practical protocol family underlying the Internet. In interviews, explain that TCP/IP collapses OSI's session/presentation/application concepts into an application layer in common representations.

8. Networking Devices

Device	Primary role	Typical layer
Hub	Repeats incoming signals to ports	Physical
Switch	Forwards frames using MAC learning	Data Link
Router	Forwards packets between networks	Network
Bridge	Connects LAN segments	Data Link
Repeater	Regenerates/repeats signals	Physical
Firewall	Allows/blocks traffic according to policy	L3/L4/L7 depending on firewall
Load balancer	Distributes traffic across backends	L4 and/or L7
Proxy	Intermediary for client traffic	Often L7

Broadcast domain

A broadcast domain is the set of devices that receive a Layer-2 broadcast. Routers normally separate broadcast domains. Switches do not automatically create separate broadcast domains unless VLANs are used.

9. Protocols and Sockets

Socket

A socket is an operating-system/application abstraction for network communication. For TCP, an established flow is identified using endpoints (commonly source IP/port and destination IP/port plus protocol).

TCP socket vs UDP socket

TCP provides a reliable ordered byte stream. UDP provides datagrams; the application decides what to do about loss, duplication or ordering.

> Interview example: A server listening on 0.0.0.0:8080 can accept connections from many clients because each established TCP connection has a distinct peer endpoint.

10. HTTP, Methods, Headers, Status Codes and Cookies

HTTP

HTTP is an application-layer request/response protocol. HTTP itself does not provide encryption; HTTPS means HTTP carried over TLS.

Common methods

Method	Typical meaning
GET	Retrieve a representation
POST	Submit data / create or trigger processing
PUT	Replace the target resource representation
PATCH	Partially modify a resource
DELETE	Remove a resource
HEAD	Like GET headers without the response body
OPTIONS	Discover supported methods/capabilities; commonly involved in CORS preflight

Status codes

Range	Meaning
1xx	Informational
2xx	Success
3xx	Redirection
4xx	Client/request error
5xx	Server-side failure
Code	Meaning
200	OK
201	Created
204	No Content
301/308	Permanent redirect variants
302/307	Temporary redirect variants
400	Bad Request
401	Authentication required/failed
403	Forbidden
404	Not Found
409	Conflict
429	Too Many Requests
500	Internal Server Error
502	Bad Gateway
503	Service Unavailable
504	Gateway Timeout

Headers

Headers carry metadata such as Host, Content-Type, Content-Length, Authorization, Cookie, Cache-Control, User-Agent and connection-related information.

Cookies

Cookies allow a browser to attach small pieces of state to matching requests. Common uses include session identifiers, preferences and security tokens. Cookie attributes such as Secure, HttpOnly and SameSite affect how cookies are transmitted and exposed to scripts.

11. How Email Works

SMTP is used primarily for sending/relaying mail. IMAP is commonly used to access and synchronize mail stored on a server. POP3 traditionally emphasizes downloading mail to a client.

Simplified path: sender client → SMTP server → recipient mail server → IMAP/POP3 client access.

> Interview trap: SMTP is not the normal protocol for reading an inbox.

12. DNS — Deep Dive

Purpose

DNS is a distributed, hierarchical naming system. It maps names to resource records such as A/AAAA addresses, MX mail servers, CNAME aliases and TXT records.

Resolution

A typical client asks a recursive resolver. If the answer is not cached, the resolver can query the hierarchy: root → TLD → authoritative server. The authoritative server returns the record; the resolver caches it according to TTL.

Example: www.shop.com → recursive resolver → .com TLD → authoritative DNS for shop.com → A/AAAA record → IP.

Record types

Record	Purpose
A	Hostname → IPv4 address
AAAA	Hostname → IPv6 address
CNAME	Alias to another hostname
MX	Mail exchange server
NS	Authoritative name servers
TXT	Arbitrary text; used for many verification/security policies
PTR	Reverse DNS mapping

DNS caching and TTL

Caching reduces latency and load. TTL controls how long a cached record may be considered fresh. A change may therefore take time to propagate through caches.

DNS over TCP/UDP

Traditional DNS commonly uses UDP for ordinary queries, while TCP is used in cases such as zone transfers and when responses require it. Modern encrypted DNS protocols include DoH and DoT; these are useful additions but are beyond Kunal's core lecture.

13. Transport Layer

The transport layer provides process-to-process communication. Ports allow multiple applications to share one host/network stack. TCP and UDP are the core protocols covered in the course.

Key ideas: segmentation/datagrams, multiplexing, reliability (TCP), flow control (TCP), congestion control (TCP), checksums, ordering and retransmission.

14. UDP

UDP is connectionless and datagram-oriented. It has low protocol overhead and does not provide TCP-style delivery, ordering or congestion-control guarantees.

Good fit	Why
DNS	Small, latency-sensitive queries; retry/fallback handled by DNS
Real-time media	Late data can be less useful than timely data
Games	Application may prioritize latency
Custom protocols	Application can implement only the reliability it needs

> Important modern nuance: HTTP/3 uses QUIC, which runs over UDP but adds reliability, congestion control, encryption and multiplexing at the QUIC layer. So 'UDP = no reliability ever' is false at the application/protocol-stack level.

15. TCP — Deep Dive

Core properties

- Connection-oriented.
- Reliable byte-stream delivery.
- Ordered delivery to the application.
- Full duplex.
- Checksum-based corruption detection.
- Retransmission of lost data.
- Flow control to protect the receiver.
- Congestion control to protect the network.

TCP header concepts

Field	Purpose
Source/Destination Port	Identify endpoints/services
Sequence Number	Position of data in the byte stream
Acknowledgment Number	Next byte expected
Flags	Connection/control state, e.g. SYN, ACK, FIN, RST
Window	Receiver-advertised flow-control capacity
Checksum	Detect corruption

Sequence numbers

TCP numbers bytes, not simply packets. If bytes 1000–1499 are sent, the next expected byte may be 1500. This enables ordering and retransmission.

16. TCP Three-Way Handshake

Connection establishment:

```
Client → Server: SYN, sequence x Server → Client: SYN + ACK, sequence y, acknowledgment x+1 Client → Server: ACK, acknowledgment y+1
```

Why three messages?

The exchange confirms reachability in both directions and synchronizes initial sequence information. It also establishes the connection state before normal data transfer.

Common interview follow-ups

- Why not two-way? The third message confirms that the client received the server's response and can communicate back.
- What if a SYN is lost? TCP retransmits according to its connection-establishment retry behavior.
- What if an ACK is lost? TCP has mechanisms that allow the connection to proceed safely when the peer retransmits information.
- What is SYN flood? An attacker sends many SYNs, consuming server resources; SYN cookies are one mitigation.

17. TCP Termination, Reliability, Flow and Congestion Control

Four-way termination

TCP is full-duplex, so each direction can be closed independently. A simplified close is FIN → ACK, FIN → ACK, though timing can vary.

RST

RST abruptly resets a connection, commonly used when a connection is invalid or a service rejects it.

Reliability

TCP uses sequence numbers, acknowledgments, checksums, timers and retransmission to provide reliable ordered delivery.

Flow control vs congestion control

Flow control	Congestion control
Protects receiver	Protects network
Uses receiver-advertised window (rwnd)	Uses congestion window (cwnd)
Question: Can the receiver keep up?	Question: Can the network keep up?

Sliding window

TCP can have multiple bytes in flight before waiting for individual acknowledgments. The effective sending window is constrained by receiver capacity and congestion state; a useful simplified expression is send_window ≈ min(rwnd, cwnd).

Congestion control

Classic TCP algorithms include slow start and congestion avoidance; real implementations use algorithms such as Reno-family variants, CUBIC and BBR. The exact algorithm is implementation-dependent.

18. Network Layer, IP and Packets

IP

IP provides best-effort, connectionless packet delivery using logical addressing and routing. It does not itself guarantee delivery, ordering or retransmission.

Packet

An IP packet contains an IP header plus payload. The payload can contain a TCP segment or UDP datagram.

TTL / Hop Limit

IPv4 TTL is decremented by routers. IPv6 uses Hop Limit. This limits the lifetime of packets and prevents infinite routing loops.

Control plane vs data plane

Control plane determines/learns routes and forwarding information. Data plane uses that information to forward packets at line rate or near line rate.

19. IPv4, IPv6, CIDR and Subnetting — FAANG Addition

IPv4 vs IPv6

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Notation	Dotted decimal	Hexadecimal with colons
Broadcast	Supported	No traditional broadcast; multicast is used
Address space	~4.3 billion	~3.4 × 10^38
Configuration	Manual/DHCP	SLAAC/DHCPv6/manual

CIDR

CIDR notation such as 192.168.1.0/24 means the first 24 bits identify the network prefix; 8 bits remain for host addressing.

Subnetting example

For a /24 IPv4 network, there are 256 addresses. In a conventional subnet, network and broadcast addresses are reserved, leaving 254 usable host addresses. Modern point-to-point and special-purpose cases have additional rules.

Interview calculation: /26 leaves 6 host bits → 2^6 = 64 addresses per subnet; conventional usable hosts = 62.

Why subnet?

Subnetting improves address organization, limits broadcast scope, enables routing boundaries and supports security/administrative segmentation.

20. Routing, Routing Tables and BGP — FAANG Addition

Routing table

A router chooses a next hop/interface based on the destination IP and routing table. Longest-prefix match is a key rule: the most specific matching prefix wins.

Example: if a router has 10.0.0.0/8 and 10.1.0.0/16, traffic to 10.1.2.3 matches both but the /16 route is more specific and wins.

Static vs dynamic routing

Static routes are manually configured. Dynamic routing protocols exchange information to adapt to topology changes.

BGP

BGP is the major inter-domain routing protocol used between autonomous systems on the Internet. It is policy-driven and exchanges reachability information rather than simply choosing the shortest physical path.

> Amazon interview reports have included BGP and its state machine in networking-focused loops, so know the purpose at a high level and understand why Internet routing is policy-based.

21. ARP, DHCP and ICMP — FAANG Addition

ARP

In IPv4 LANs, ARP resolves an IPv4 address to a link-layer MAC address on the local network. Example: host knows 192.168.1.20 but needs the Ethernet destination MAC. It broadcasts an ARP request and receives a reply.

DHCP

DHCP automatically provides network configuration such as IP address, subnet mask, default gateway and DNS server.

DORA: Discover → Offer → Request → Acknowledge.

ICMP

ICMP carries network-control and diagnostic messages. ping uses ICMP Echo Request/Reply; traceroute commonly relies on TTL/Hop-Limit expiry and ICMP responses (implementation varies).

> These three topics have appeared in Amazon networking interview reports and are useful for troubleshooting questions.

22. NAT and Middleboxes

NAT

Network Address Translation rewrites addresses and often ports as traffic crosses a boundary. A common home-NAT pattern lets many private hosts share one public IPv4 address.

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Example: Laptop 192.168.1.10:52000 → Router NAT → Public 203.0.113.5:40001 → Server:443
```

PAT

Port Address Translation is a common form of NAT in which the router uses source ports to distinguish multiple internal connections sharing one public IP.

Middleboxes

Firewalls, NAT gateways, proxies, load balancers and IDS/IPS can inspect, rewrite, allow, deny or route traffic.

Interview trap

> NAT is not a replacement for a firewall. NAT can reduce direct reachability, but security policy should be enforced explicitly by firewall/security controls.

23. MTU, MSS and Fragmentation — Amazon Interview Topic

MTU

Maximum Transmission Unit is the largest IP packet payload a link can carry without fragmentation at that link's IP layer. Ethernet commonly uses an MTU of 1500 bytes, but the exact value depends on the network.

MSS

Maximum Segment Size is a TCP concept describing the maximum TCP payload size a host is willing to receive in a segment, excluding TCP/IP headers. A common Ethernet/IPv4 example is MSS 1460 = 1500 MTU – 20-byte IPv4 header – 20-byte TCP header. Options can change the exact relationship.

MTU vs MSS

MTU	MSS
IP-layer/link property	TCP-layer payload limit
Includes IP header in the packet size	Excludes TCP/IP headers; refers to TCP data
Applies beyond TCP	Only TCP

> Amazon interview reports have explicitly included 'explain MTU and compare MSS and MTU'.

24. HTTP vs HTTPS and TLS — FAANG Addition

HTTPS

HTTPS is HTTP carried over TLS. TLS provides confidentiality, integrity and server authentication when certificate validation succeeds.

High-level TLS flow

1. Client sends a ClientHello with supported versions/ciphers and key-share information.
2. Server responds with selected parameters, certificate and key-share information.
3. Both sides derive shared traffic keys using the negotiated cryptographic exchange.
4. After the handshake, application data is encrypted and integrity-protected.

Why certificates?

Certificates bind a public key to an identity under a trust model. The client validates the certificate chain, hostname and validity conditions before trusting the server identity.

TLS vs TCP

TCP is a transport protocol. TLS is a security protocol that can run above a transport. They solve different problems.

> Amazon interview reports include questions on HTTPS, SSL/TLS and handshake concepts.

25. HTTP/1.1 vs HTTP/2 vs HTTP/3 and QUIC — FAANG Addition

Feature	HTTP/1.1	HTTP/2	HTTP/3
Transport	TCP	TCP	QUIC over UDP
Framing	Text-oriented protocol semantics	Binary frames	QUIC streams + HTTP/3 frames
Multiplexing	Limited; multiple requests often contend per connection	Multiplexed streams	Multiplexed streams
Header compression	No HPACK	HPACK	QPACK
TLS	Typically TLS for HTTPS	TLS for HTTPS in practice	TLS 1.3 integrated into QUIC

Head-of-line blocking

TCP delivers one ordered byte stream. If an early packet is lost, later bytes cannot be delivered to the application as if the missing bytes did not exist. HTTP/2 multiplexes streams over TCP, so packet loss can still affect the connection. QUIC uses independent streams over UDP, reducing cross-stream head-of-line blocking.

Why HTTP/3 uses UDP

QUIC builds reliable, encrypted, multiplexed transport behavior over UDP, allowing protocol evolution without requiring a new kernel TCP implementation.

26. Proxies, Reverse Proxies and Load Balancers

Forward proxy

A forward proxy acts on behalf of clients. It can enforce policy, cache content or hide client network details from the destination.

Reverse proxy

A reverse proxy sits in front of servers. It can terminate TLS, route requests, cache responses, enforce limits and protect/hide backend topology.

Load balancer

A load balancer distributes traffic across multiple backend instances. L4 load balancers use transport-level information; L7 load balancers understand application protocols such as HTTP.

```
Example: Client → ALB/reverse proxy → service instances A/B/C → database/cache.
```

27. CDN, Caching and Anycast — FAANG Addition

CDN

A Content Delivery Network places edge servers geographically closer to users and caches eligible content. This reduces latency and origin load.

Cache hit/miss

Cache hit: edge already has a valid object. Cache miss: edge fetches from origin or another upstream, stores it if policy allows, and serves the client.

Anycast

Anycast allows multiple locations to advertise the same IP prefix; routing directs a client toward a topologically appropriate location. It is widely used for resilient distributed services and DNS infrastructure.

28. WebSockets and Long Polling — FAANG Addition

Long polling

The client sends a request and the server holds it until data is available or a timeout occurs. The client then reconnects.

WebSocket

WebSocket provides a persistent, full-duplex channel after an HTTP-based upgrade handshake. It is useful for chat, collaborative apps, live dashboards and other real-time interactions.

SSE

Server-Sent Events provide a one-way server-to-browser stream over HTTP. Use them when the server mainly pushes events and the client does not need a full-duplex channel.

29. Firewall, VPN and Security Basics

Firewall

A firewall applies allow/deny policies to traffic. It may inspect IPs, ports, connection state or application data depending on its type.

VPN

A VPN creates an authenticated/encrypted tunnel across an untrusted network. The exact behavior depends on the VPN protocol and architecture.

Common interview security concepts

- Confidentiality → encryption.
- Integrity → cryptographic integrity/authentication mechanisms.

- Authentication → proving identity.
- Authorization → deciding what an authenticated identity may do.
- Least privilege → grant only required access.
- TLS certificate validation → protects against impersonation when the trust chain is correctly validated.

30. Latency, Bandwidth, Throughput, Jitter and Packet Loss

Term	Meaning
Latency	Time for data/response to travel; often measured round-trip for network tests
Bandwidth	Maximum capacity/rate of a link
Throughput	Actual achieved data rate
Jitter	Variation in packet delay
Packet loss	Packets that fail to reach the intended destination

Key distinction

A high-bandwidth link can still have high latency. Bandwidth answers 'how much per unit time can the link carry'; latency answers 'how long does a given packet take to get there'.

Example

A large file transfer benefits from bandwidth, while an interactive game or database request is often very sensitive to latency and jitter.

31. End-to-End: What Happens When You Type amazon.com?

This is a high-value SDE-1 interview question because it tests whether you can connect many networking concepts into one causal chain.

1. The browser parses the URL and checks local/browser caches and connection state.
2. The hostname is resolved using DNS. A cached result may avoid a full DNS hierarchy lookup.
3. The client chooses a destination IP and transport protocol. For HTTPS over HTTP/1.1 or HTTP/2 this commonly means TCP; HTTP/3 uses QUIC over UDP.
4. If TCP is used and no suitable connection exists, TCP performs the three-way handshake.
5. For HTTPS over TCP, TLS negotiation establishes encrypted application traffic and validates the server certificate.
6. The browser sends the HTTP request with headers such as Host and relevant cookies.
7. Packets/frames travel through local Wi■Fi/Ethernet, the gateway/router, NAT or security middleboxes, and multiple routed networks.
8. The server-side load balancer/reverse proxy may select a backend.
9. The application processes the request, possibly reading caches/databases/services.
10. The response travels back. TCP acknowledges/retransmits as needed; QUIC performs equivalent transport reliability at its layer.
11. The browser parses HTML, discovers subresources, makes additional requests, and renders the page.

Strong Amazon-style answer: explicitly state that the exact path depends on protocol version, cache state, existing connections, CDN/load balancer architecture and network conditions. Do not claim every request always performs a fresh

DNS lookup, TCP handshake and TLS handshake.

32. Troubleshooting by Layer

Use layered reasoning rather than guessing.

Layer	Example failure	Useful question/tool
Application	500/404, bad API response	Logs, curl, browser devtools
Transport	Connection refused/timeout	ss/netstat, connection tests
Network	No route, wrong gateway	ip route, traceroute
Data Link	ARP/MAC/VLAN issue	arp/ip neigh, switch/VLAN checks
Physical	Cable/radio/link down	Interface/link status

Useful tools

- ping — reachability/latency using ICMP or platform equivalent.
- traceroute/tracert — path diagnosis using TTL/Hop Limit behavior.
- nslookup/dig — DNS queries.
- curl — HTTP/API behavior.
- ss/netstat — sockets and listening ports.
- tcpdump/Wireshark — packet-level inspection.
- ip addr / ip route — interface addresses and routing tables on Linux.

33. Amazon / FAANG Interview Questions with Model Answers

Q1. TCP vs UDP?

TCP is connection-oriented, reliable and ordered, with retransmission, flow control and congestion control. UDP is connectionless and datagram-oriented with lower overhead and no TCP-style delivery/ordering guarantees. Choose TCP when correctness and reliable ordering matter; choose UDP when low overhead/latency matters or when a higher-level protocol such as QUIC supplies the needed reliability.

Q2. Explain the TCP three-way handshake.

SYN establishes the client's intent and initial sequence information; SYN-ACK acknowledges it and supplies the server's sequence information; ACK confirms receipt. Both sides then have synchronized connection state.

Q3. What happens when you type amazon.com?

DNS resolves the hostname; the client selects a destination and transport; TCP may handshake or an existing connection may be reused; TLS may negotiate for HTTPS; HTTP request/response travels through routers, NAT, security devices and possibly CDN/load balancers; the browser receives and renders the response.

Q4. What is DNS?

A distributed hierarchical naming system. A recursive resolver obtains records from cache or authoritative DNS infrastructure. Common records include A, AAAA, CNAME and MX.

Q5. Router vs switch?

A switch primarily forwards Layer-2 frames using MAC addresses inside a LAN/VLAN. A router forwards Layer-3 packets between networks using IP and routing information.

Q6. What is NAT?

NAT rewrites IP addresses and often ports at a network boundary. A common home router maps multiple private clients to one public IPv4 address using port translations.

Q7. MTU vs MSS?

MTU is the maximum IP packet size supported by a link without fragmentation at that link. MSS is the maximum TCP payload size a receiver advertises. MSS is therefore smaller than the MTU after accounting for IP and TCP headers in common cases.

Q8. Flow control vs congestion control?

Flow control protects the receiver; TCP's receive window tells the sender how much the receiver can accept. Congestion control protects the network; the congestion window changes based on perceived network conditions.

Q9. Why can HTTP/2 still suffer from head-of-line blocking?

HTTP/2 multiplexes streams over one TCP connection, but TCP itself is an ordered byte stream. Loss of an earlier TCP segment can prevent later bytes from being delivered to the application. QUIC's independent streams reduce this cross-stream effect.

Q10. What is HTTPS?

HTTP carried over TLS. TLS provides encryption, integrity protection and server authentication through certificate validation.

Q11. What is a socket?

An OS/application communication endpoint. For TCP, an established connection is identified by endpoint information including source/destination IPs and ports plus the transport protocol.

Q12. What is a subnet?

A logical subdivision of an IP network. CIDR notation such as /24 defines the number of network-prefix bits and helps routers and hosts determine which destinations are local or remote.

Q13. What is DHCP?

A protocol for dynamically supplying network configuration. The classic DORA sequence is Discover, Offer, Request, Acknowledge.

Q14. What is ARP?

IPv4 LAN address resolution: mapping a known local IPv4 address to a MAC address so the host can send an Ethernet frame.

Q15. What is BGP?

The main inter-domain routing protocol of the Internet. It exchanges reachability information between autonomous systems and makes policy-based routing decisions.

Q16. What is a reverse proxy?

A server-side intermediary in front of backends. It can terminate TLS, route requests, cache, enforce policy and hide backend topology.

Q17. Why is UDP used by HTTP/3?

HTTP/3 uses QUIC, which runs over UDP. QUIC implements reliable, encrypted, multiplexed transport in user space and avoids some limitations of TCP's stream-level head-of-line blocking.

Q18. What is TTL?

IPv4 TTL and IPv6 Hop Limit constrain how many router hops a packet may survive. Routers decrement the value; expiry causes the packet to be discarded and commonly generates an ICMP diagnostic response.

Q19. What is a broadcast domain?

The set of Layer-2 endpoints that receive a broadcast. Routers separate broadcast domains; VLANs can split a switch infrastructure into separate Layer-2 broadcast domains.

Q20. How would you debug 'API is timing out'?

Start at the application and move down: confirm DNS resolution, test TCP connection to the correct host/port, inspect routes/firewalls/security groups, capture packets if needed, verify server accept/listen state, then inspect application/backend latency. Distinguish connection timeout from server-processing timeout.

34. Rapid Revision Cheat Sheets

OSI in 30 seconds

7 Application — HTTP/DNS/SMTP 6 Presentation — representation/encryption/compression concepts 5 Session — session management 4 Transport — TCP/UDP/ports 3 Network — IP/routing 2 Data Link — MAC/frames/Ethernet/Wi■Fi 1 Physical — bits/signals

TCP vs UDP

TCP	UDP
Connection-oriented	Connectionless
Reliable ordered byte stream	Best-effort datagrams
ACK/retransmission	No TCP-style ACK/retransmission
Flow + congestion control	No TCP-style flow/congestion control
More overhead	Lower overhead
HTTP/1.1, HTTP/2, SSH, etc.	DNS, media/games; QUIC runs over UDP

Numbers to remember

Fact	Value
IPv4	32 bits
IPv6	128 bits
Port	16 bits, 0–65535
HTTP	80 commonly
HTTPS	443 commonly
DNS	53
SSH	22
Typical Ethernet MTU	1500 bytes (common, not universal)

Golden mental model

Application data → transport segment/datagram → IP packet → link-layer frame → bits → network → reverse process at destination.

Amazon SDE-1 priority order

1. TCP vs UDP + handshake + reliability.
2. DNS resolution and caching.
3. HTTP/HTTPS + TLS.
4. OSI/TCP-IP and devices.
5. IP addressing, subnetting and CIDR.
6. NAT, ports and sockets.
7. Flow control vs congestion control.
8. MTU/MSS and fragmentation.
9. ARP/DHCP/ICMP.
10. Routing/BGP and troubleshooting.
11. HTTP/2, HTTP/3/QUIC, CDN, load balancers and WebSockets.

Study rule: For every topic, practice answering four things: (1) what it is, (2) why it exists, (3) how it works, and (4) one concrete example/trade-off. This is much closer to SDE interview performance than memorizing definitions.

Sources and Scope

Primary course: Kunal Kushwaha, “Computer Networking Full Course - OSI Model Deep Dive with Real Life Examples,” YouTube. The published syllabus/timestamps cover the course topics from networking fundamentals through OSI/TCP-IP, HTTP, DNS, TCP/UDP, IP, IPv4/IPv6, middleboxes, NAT and the data-link layer.

This PDF adds interview-focused material that is not fully covered in the course, including CIDR/subnetting, ARP, DHCP, ICMP, BGP, MTU/MSS, TLS/HTTPS, HTTP/2/HTTP/3, QUIC, CDN, load balancing, WebSockets and layered troubleshooting.

Interview reports used to prioritize additions include publicly reported Amazon interviews mentioning URL-request flow, HTTP/HTTPS/TLS, router/switch/hub, MTU/MSS, TCP/UDP, flow/error control, public IP, DHCP, DNS, VPN, subnetting, BGP and OSI topics. Interview reports are anecdotal and should be treated as preparation signals, not guarantees of what any specific Amazon interviewer will ask.

The notes are intentionally original and paraphrased. They do not reproduce the video transcript.

Recommended official references for deeper protocol details: RFC 9293 (TCP), RFC 8200 (IPv6), RFC 9110 (HTTP semantics), RFC 8446 (TLS 1.3), RFC 9000 (QUIC), RFC 1034/1035 (DNS), RFC 2131 (DHCP), RFC 826 (ARP), RFC 4271 (BGP).